

ESBJERG AP, Denmark

WMO: 060800

Lat: 55.528N

Lon: 8.563E

Elev: 97

StdP: 14.64

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.3	21.1	12.3	10.3	20.4	16.2	12.5	23.3	34.9	43.8	31.5	43.6	7.0	50	0.653

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.8	80.3	65.6	76.7	64.3	73.1	62.6	67.5	75.5	65.6	73.1	63.9	70.5	9.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	91.4	71.0	62.8	85.8	68.6	61.2	80.9	66.7	31.9	75.8	30.5	73.6	29.3	70.9	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.6	24.4	21.7	DB	12.8	84.9	6.4	3.6	8.3	87.5	4.5	89.6	1.0	91.6	-3.6	94.2	(4)
(5)			WB	12.1	71.1	6.2	2.1	7.7	72.6	4.1	73.8	0.7	74.9	-3.8	76.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.0	34.8	35.2	38.6	45.8	52.4	57.3	61.8	62.1	57.0	50.0	42.6	37.6	(6)	
	DBStd	11.35	6.90	6.65	5.64	6.00	5.92	4.95	4.88	4.54	4.47	5.37	6.11	7.18	(7)	
	HDD50	2116	471	413	353	155	40	2	0	0	3	67	228	384	(8)	
	HDD65	6265	936	833	818	577	394	237	128	115	242	465	671	849	(9)	
	CDD50	1389	0	0	0	28	113	222	366	374	213	67	6	0	(10)	
	CDD65	66	0	0	0	0	2	7	29	25	3	0	0	0	(11)	
	CDH74	553	0	0	0	1	22	75	250	192	13	0	0	0	(12)	
	CDH80	88	0	0	0	0	1	6	44	36	1	0	0	0	(13)	
(14) Wind	WSAvg	10.6	11.7	12.2	12.2	10.5	10.2	10.3	9.2	8.9	9.8	10.4	10.6	11.1	(14)	
(15) Precipitation	PrecAvg	35.0	3.1	2.3	2.0	1.7	1.9	2.6	2.8	3.5	3.7	4.0	3.5	3.6	(15)	
	PrecMax	43.9	5.4	5.0	4.1	3.5	4.1	5.1	4.9	6.9	8.0	9.8	7.0	7.2	(16)	
	PrecMin	23.0	0.1	0.6	0.3	0.2	0.1	0.0	0.6	0.8	0.9	1.3	0.9	0.8	(17)	
	PrecStd	5.4	1.4	1.1	1.1	0.8	0.9	1.3	1.3	1.6	1.7	1.7	1.6	1.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.6	48.3	57.1	70.2	76.9	80.6	84.4	84.2	75.6	64.4	57.0	51.6	(19)	
		MCWB	47.0	46.1	48.9	56.2	63.3	64.8	66.7	64.6	64.9	59.8	54.9	49.6	(20)	
	2%	DB	46.7	46.4	51.6	65.3	72.0	75.6	80.5	80.2	70.7	61.0	54.8	49.8	(21)	
		MCWB	45.4	45.2	46.1	53.5	59.1	62.8	66.4	65.3	62.5	57.8	52.7	48.9	(22)	
	5%	DB	45.1	44.8	48.2	60.8	68.0	71.6	76.8	75.2	66.6	59.1	52.6	48.2	(23)	
		MCWB	43.8	43.7	44.9	51.8	57.1	61.5	64.5	64.4	60.6	56.3	50.4	47.0	(24)	
	10%	DB	44.2	43.2	46.1	55.7	64.0	67.6	72.0	71.4	64.3	57.4	50.6	46.5	(25)	
		MCWB	42.9	42.2	43.7	49.3	55.3	59.6	62.9	63.1	59.3	54.9	48.6	45.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.5	46.5	49.9	59.0	65.2	68.0	70.2	70.0	67.0	60.3	55.4	50.1	(27)	
		MCDB	48.4	47.2	53.5	67.2	74.3	75.9	79.8	77.6	72.8	63.0	56.6	50.7	(28)	
	2%	WB	45.6	45.0	47.0	54.8	60.9	65.0	67.8	67.1	63.7	58.7	53.1	48.6	(29)	
		MCDB	46.3	45.6	49.9	62.6	69.0	73.0	76.9	74.0	68.4	60.7	54.7	49.1	(30)	
	5%	WB	43.9	43.8	45.2	52.4	58.6	62.4	65.7	65.3	61.9	57.1	51.0	46.8	(31)	
		MCDB	44.7	44.4	47.0	59.0	65.6	69.3	73.6	72.4	65.7	58.8	52.5	47.5	(32)	
	10%	WB	42.3	42.5	43.8	50.2	56.1	60.1	63.8	63.8	60.2	55.5	49.1	45.3	(33)	
		MCDB	43.2	43.2	45.2	55.1	62.5	66.1	70.5	70.1	63.3	57.5	50.4	46.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	8.1	10.8	14.9	15.2	14.1	14.8	14.8	12.5	10.2	8.3	7.7	(35)	
		MCDBR	6.3	7.8	14.3	21.9	22.7	22.7	23.0	21.7	16.6	10.9	7.2	6.5	(36)	
	5% WB	MCWBR	6.6	6.8	10.2	12.8	12.9	12.9	11.6	10.3	10.3	8.7	7.1	6.7	(37)	
		MCWBR	6.2	7.1	11.5	19.1	19.4	19.4	19.9	17.8	14.1	9.8	7.1	6.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.323	0.340	0.362	0.395	0.386	0.383	0.396	0.398	0.378	0.357	0.333	0.310	(40)	
		taud	2.247	2.298	2.299	2.257	2.343	2.373	2.351	2.355	2.396	2.381	2.302	2.245	(41)	
	Ebn at Noon		183	229	253	261	272	273	266	258	247	221	179	159	(42)	
		Edh at Noon	18	25	31	37	36	36	36	34	29	23	17	15	(43)	
(44) All-Sky Solar Radiation	RadAvg		156	351	770	1285	1678	1773	1654	1323	915	474	193	115	(44)	
	RadStd		27	48	71	151	165	138	175	95	95	57	30	12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-202	-288	N/A	N/A

Nomenclature: See separate page